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OncoHealth Assessment Interim Readout

October 8, 2025

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The Newfire team, with help from engaged OncoHealth team members, is wrapping up the technical investigations, and starting to prepare our findings and recommendations in their final form. This document highlights the findings and recommendations to date.

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1 - Review of Technical Findings

- **Matis UM core -** We found pervasive technical debt throughout the Matis UI, monolith, and OADB - with evidence of 2x to 10x load on client implementations, Change Requests, and new development (e.g., Monarch), with reputational and liability risks due to mishandling cases, including erroneous approval of multi-million dollar drug regimens.

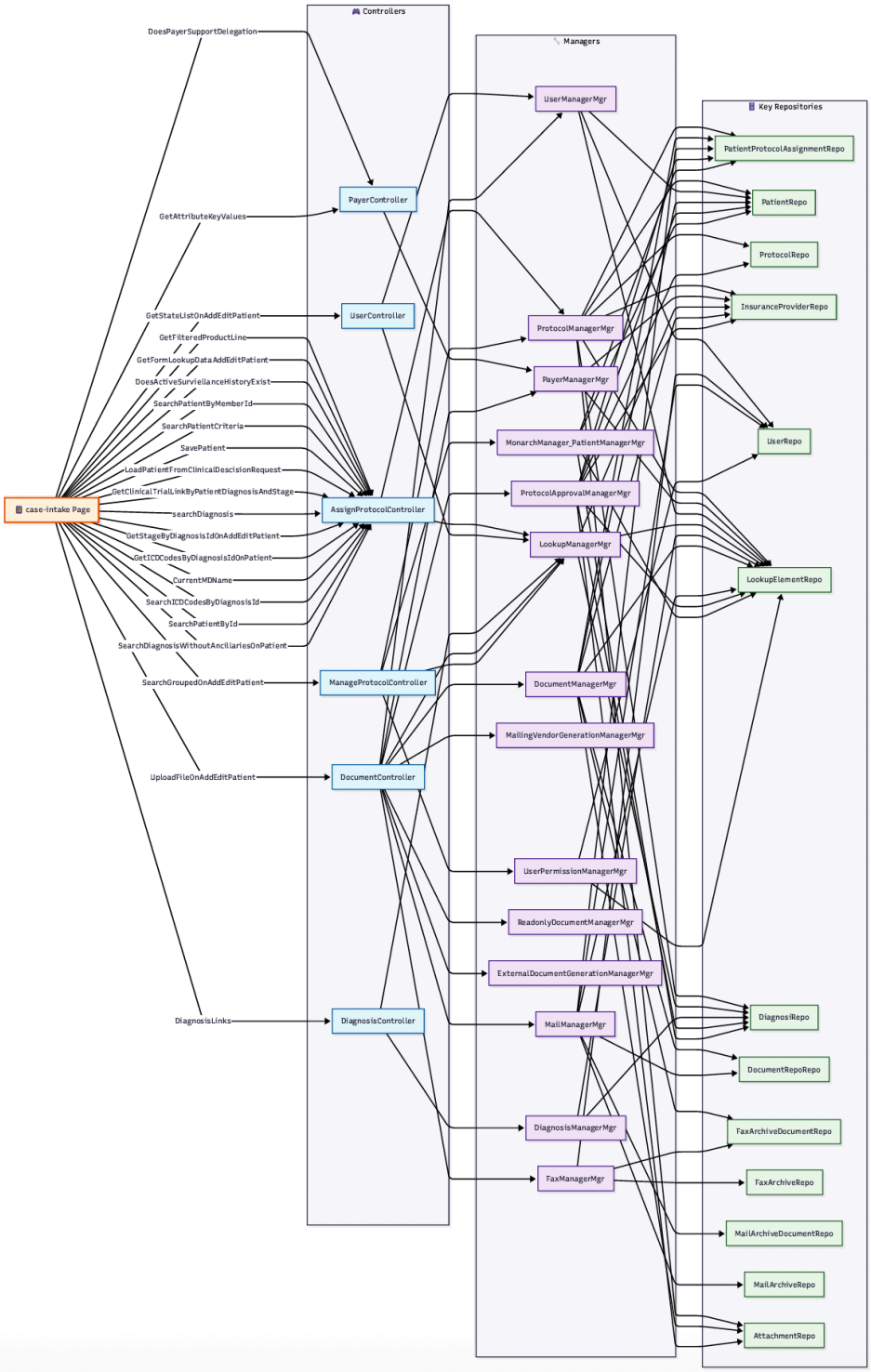
  This technical debt has shaped the engineering organization's structure in a way that perpetuates the inefficiencies in the core system. The workstreams for most of the development teams are filled with client implementation development tasks, often incorporating teams that were intended to focus on strategic projects and optimizations. Selective refactoring is not considered a viable option, so almost all new work adds to the existing debt, QA, and D&A loads.

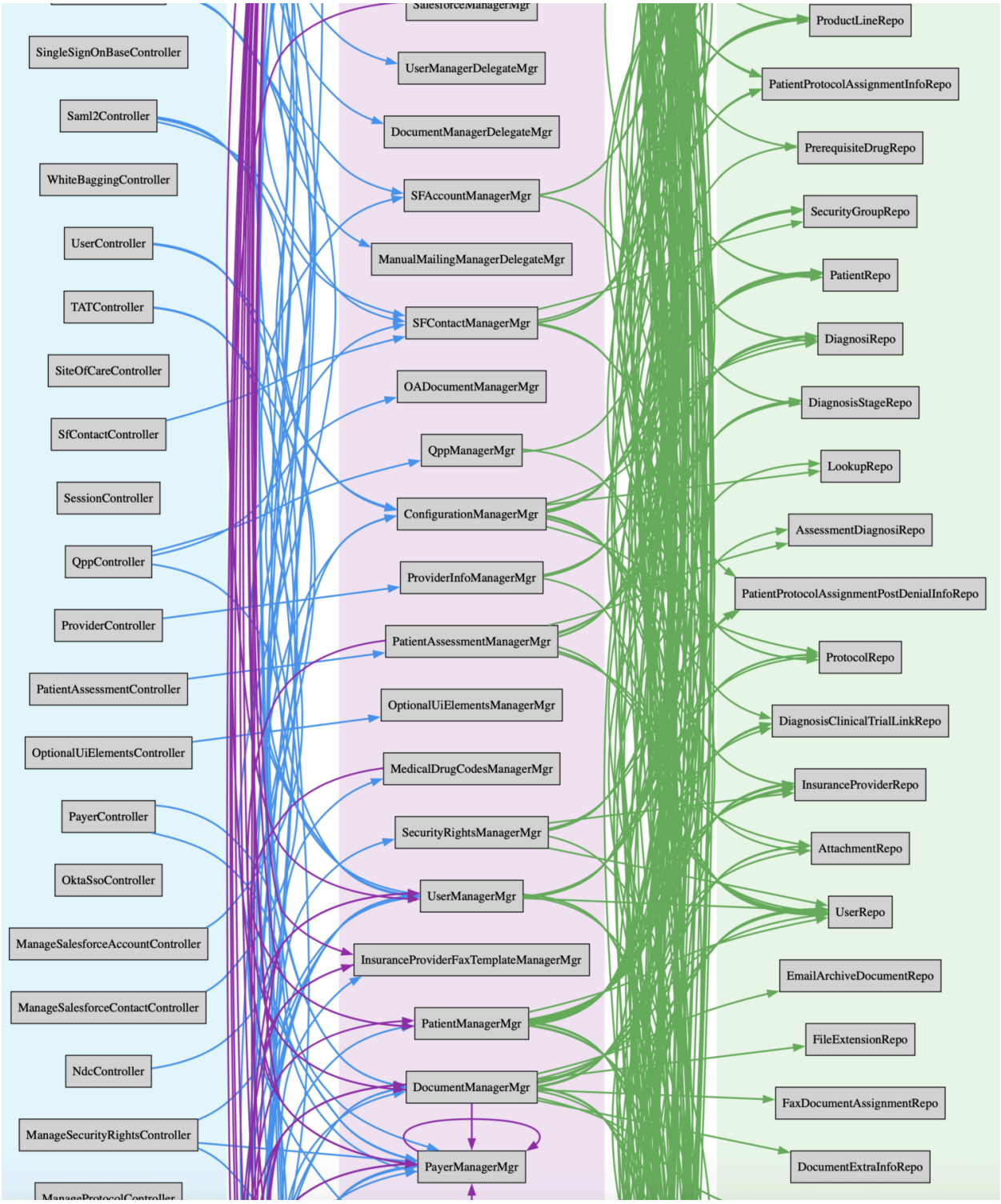
  This technical debt is mostly found in the Matis core. The other constituent services in OneUM architecture are either operating well, or have their own findings that can be addressed in separate projects. See the bullet for "OneUM Service Architecture" below in this section for more details.

- **Matis Monolith** - Has deep complexity, mostly in the form of hard-coded case workflow. This drives the multiplier on scope for any work that touches it. Matis doesn't have viable configurability. A code quality review highlighted additional critical issues.

  The following two diagrams show the system mappings and the high-level of complexity.
  - The first diagram shows the controllers, managers and data repos that are called from a single page in the Matis UI. The page shown is the first case intake page, up to but not including following steps for protocol assignment.
  - The second diagram shows a zoom-in on the total set of controller-manager-repo mappings. This effectively maps groupings of endpoint APIs to database tables. Of note here is how many manager-manager calls there are, shown in purple.

  NOTE: the embedded versions of these documents are provided to demonstrate scope and complexity. For human-readable versions of these please see supporting materials (to be uploaded on SharePoint). These diagrams are generated by a custom script and we can produce specific slices on-demand.


Controllers
Managers
Key Repositories
DoesPayerSupportDelegation
GetAttributeKeyValues
GetStateListOnAddEditPatient
GetFilteredProductLine
GetFormLookupDataAddEditPatient
DoesActiveSurviellanceHistoryExist
SearchPatientByMemberId
SearchPatientCriteria
SavePatient
LoadPatientFromClinicalDescisionRequest
GetClinicalTrialLinkByPatientDiagnosisAndStage
searchDiagnosis
GetStageByDiagnosisIdOnAddEditPatient
GetICDCodesByDiagnosisIdOnPatient
CurrentMDName
SearchICDCodesByDiagnosisId
SearchPatientById
SearchDiagnosisWithoutAncillariesOnPatient
SearchGroupedOnAddEditPatient
UploadFileOnAddEditPatient
DiagnosisLinks
case-intake Page
PayerController
UserController
AssignProtocolController
ManageProtocolController
DocumentController
DiagnosisController
UserManagerMgr
ProtocolManagerMgr
PayerManagerMgr
MonarchManager_PatientManagerMgr
ProtocolApprovalManagerMgr
LookupManagerMgr
DocumentManagerMgr
MailingVendorGenerationManagerMgr
UserPermissionManagerMgr
ReadonlyDocumentManagerMgr
ExternalDocumentGenerationManagerMgr
MailManagerMgr
DiagnosisManagerMgr
FaxManagerMgr
PatientProtocolAssignmentRepo
PatientRepo
ProtocolRepo
InsuranceProviderRepo
UserRepo
LookupElementRepo
DiagnosisRepo
DocumentRepoRepo
FaxArchiveDocumentRepo
FaxArchiveRepo
MailArchiveDocumentRepo
MailArchiveRepo
AttachmentRepo


SingleSignOnBaseController
Saml2Controller
WhiteBaggingController
UserController
TATController
SiteOfCareController
SfContactController
SessionController
QppController
ProviderController
PatientAssessmentController
OptionalUiElementsController
PayerController
OktaSsoController
ManageSalesforceAccountController
ManageSalesforceContactController
NdcController
ManageSecurityRightsController
UserManagerDelegateMgr
DocumentManagerDelegateMgr
SFAccountManagerMgr
ManualMailingManagerDelegateMgr
SFContactManagerMgr
OADocumentManagerMgr
QppManagerMgr
ConfigurationManagerMgr
ProviderInfoManagerMgr
PatientAssessmentManagerMgr
OptionalUiElementsManagerMgr
MedicalDrugCodesManagerMgr
SecurityRightsManagerMgr
UserManagerMgr
InsuranceProviderFaxTemplateManagerMgr
PatientManagerMgr
DocumentManagerMgr
PayerManagerMgr
ProductLineRepo
PatientProtocolAssignmentInfoRepo
PrerequisiteDrugRepo
SecurityGroupRepo
PatientRepo
DiagnosiRepo
DiagnosisStageRepo
LookupRepo
AssessmentDiagnosiRepo
PatientProtocolAssignmentPostDenialInfoRepo
ProtocolRepo
DiagnosisClinicalTrialLinkRepo
InsuranceProviderRepo
AttachmentRepo
UserRepo
EmailArchiveDocumentRepo
FileExtensionRepo
FaxDocumentAssignmentRepo
DocumentExtraInfoRepo

- **Matis OADB** - The OADB schema is overgrown and opaque for new developers. It contains a large number of disused tables and columns which complicate reporting and developer onboarding. New clients and CRs are often developed on top of the existing schema without factoring in prior patterns.

  Critically, the OADB schema contains many *Overloaded* columns, which have different semantic meanings depending on the customer. This “eroded” schema design reflects the level of custom coding deployed on an implementation by implementation basis.

  OADB’s transactional design does not provide standard metadata for tracing and auditing. It does not cleanly support “effective as of this date” information concepts. The system also lacks standardized database version control, which complicates testing and deployments. Documentation is minimal.

  The following page is the current map of the Matis OADB. Note that some of the tables are disused and we are working with Onco DevOps to retrieve last-updated dates for each table from the Production database. These will be noted in our final deliverables.

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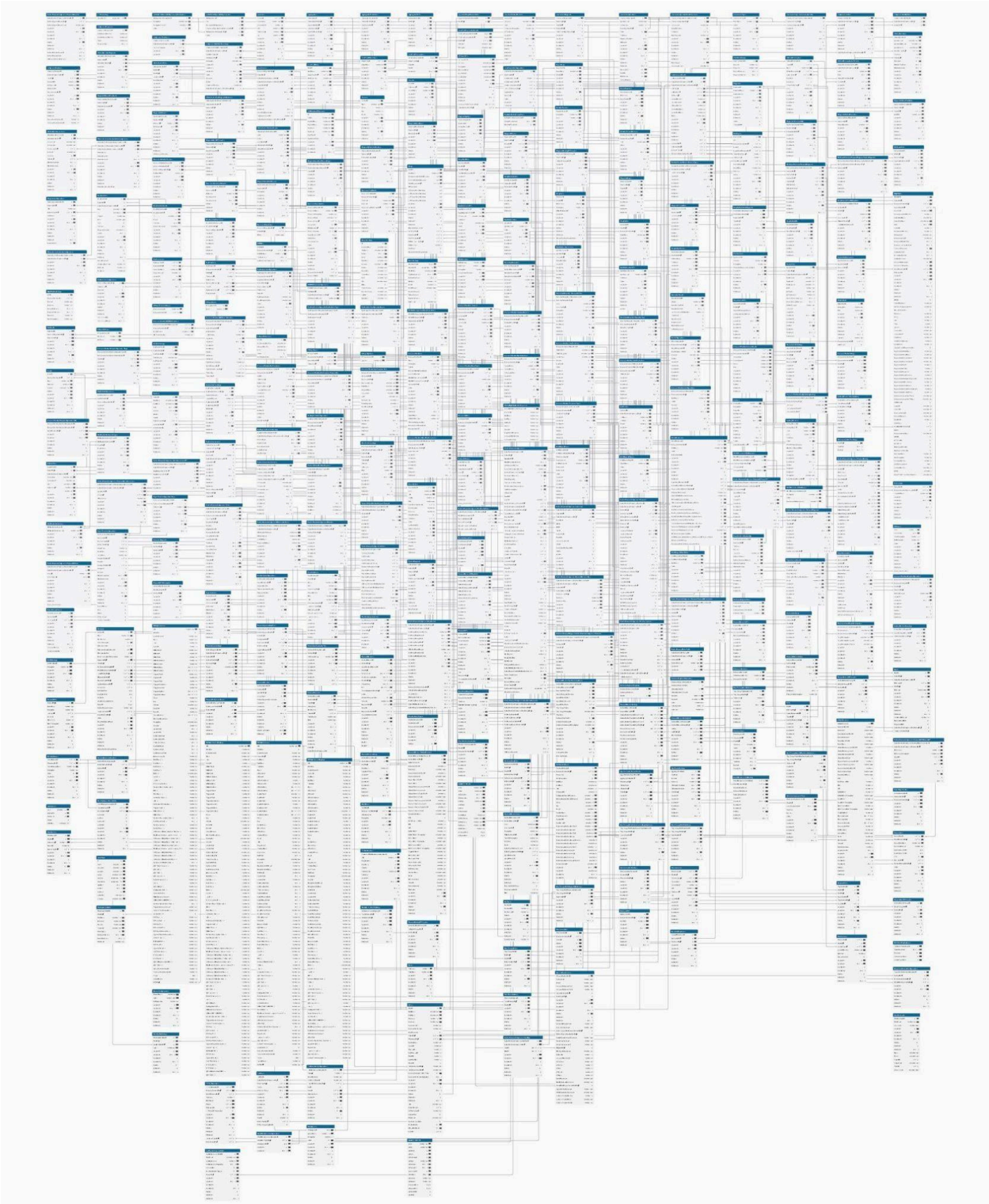
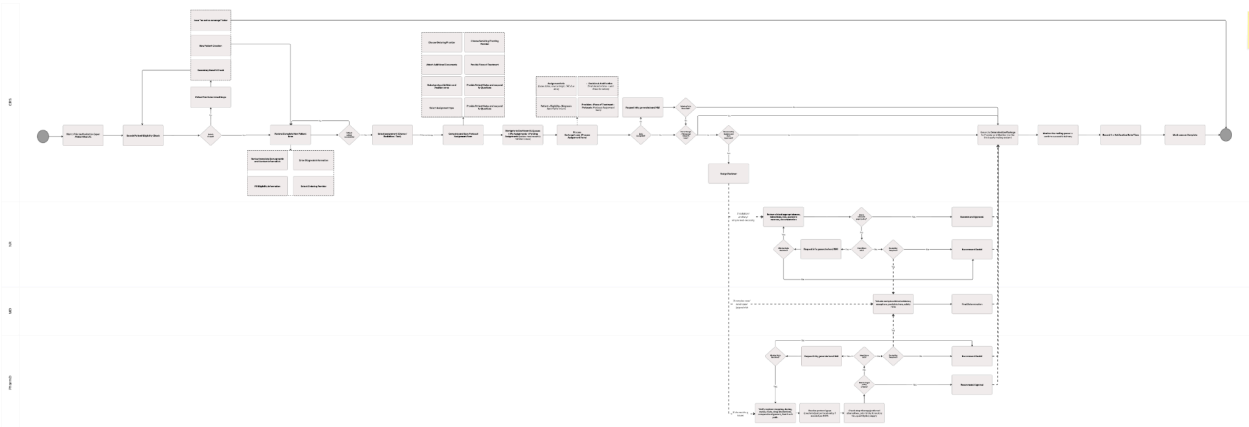


- **Matis UI** - Consists roughly of 20 main pages but built on 3 UI architectures (Angular v1 and Angular v18 single-page applications, embedded in .NET Razor pages that generate additional UI).

  The UI creates business, reporting and technology challenges, including:

  - The approach requires new, hard-coded flows for each client and project, to match hard-coded business logic on the server..
  - The user experience provides warnings for data entry issues,but does not enforce them. This allows users to enter data that can (and does) cause downstream reporting issues.
  - The user experience itself creates significant operations drag.

  A generic case workflow illustrated below - it does not account for client-specific customizations, such as post-decision workflows.



- **Monarch** (Admin Portal and DrugsDB integrated with Matis) - Monarch was conceived as a replacement of the Matis monolith, with improved schemas for drugs and providers. We understand that it was de-scoped a couple times and that forced a model where the Matis monolith remained, and new services had to integrate with it and work with its complexities.
  - There was good work done on the Drugs db and Admin Portal that are integrated with SmartUM. We believe the Drugs service that was built for the integration with Matis can also be used in future work.

    - A new provider schema was proposed and it looks feasible as a starting point for future work.
    - Integration with the hard-coded and conflicting Matis workflows added complexity to the Monarch project. Feature flags were built to break the work on a client-by-client basis, but each client-specific implementation still bore a large amount of development and QA time.
    - To enable client-by-client implementations, Monarch data had to be integrated back into Matis clients. This was identified fairly recently and drove significant scope on back-integrating DrugsMS schema into Matis for both medical and pharma clients.

- **SmartUM** - Five payers were migrated from Matis to SmartUM. New clients have never been implemented directly on SmartUM. It is relatively stable for the existing clients, but the functionality is limited and has driven redundant development. This includes, but is not limited to redundant TaT modules and redundant D&A reporting.
    - As Matis is improved SmartUM is falling behind (TaTs, reporting improvements, etc).
    - HealthFortis has been slow to respond to change requests and makes changes without notice.
    - Partial access to SmartUM data - 27 tables downloaded daily. Unknown ownership/access to case workflow metadata. Full data has never been shared.
        - We recommend a follow up on contractual rights to data.
    - Future migrations out of SmartUM look to have significant challenges in Data Migration, reimplementation of case workflows, and migration of redundant services. This poses a risk for OncoHealth for the clients on SmartUM.
    - *SmartUM has a potential business continuity risk. We recommend taking steps to ensure a feasible migration path, and that agreements are put in place to receive data in a usable format on request or in the case of a HealthFortis failure.*
    - *For the above reasons we currently would caution against implementing new clients on SmartUM, versus Matis. In addition:*
        - Tradeoffs have been documented on both sides but it appears that SmartUM will generate more work in any future platform consolidation.
        - SmartUM will struggle to stay in parity with Matis/NewUM Improvements
        - SmartUM has a lower load, today, on the clinical team, but not in other areas of compliance and reporting.
        - Built on a Java back-end with GWT (Java-based) front-end.
        - Releases are manually-deployed to VMs hosted in Otava.

- CI/CD systems are best practice for automated deployments. Without that, the team has experienced infrequent back-end releases, and human error.
- Otava is a mid-tier healthcare-aware hosting service, but will not have the reliability and breadth of service as ADO or AWS.

- **OneUM Service Architecture** - The overall service architecture on Azure is workable and many of the services and 3rd party integrations are good-enough to support the business. The following diagram highlights services based on business risk.

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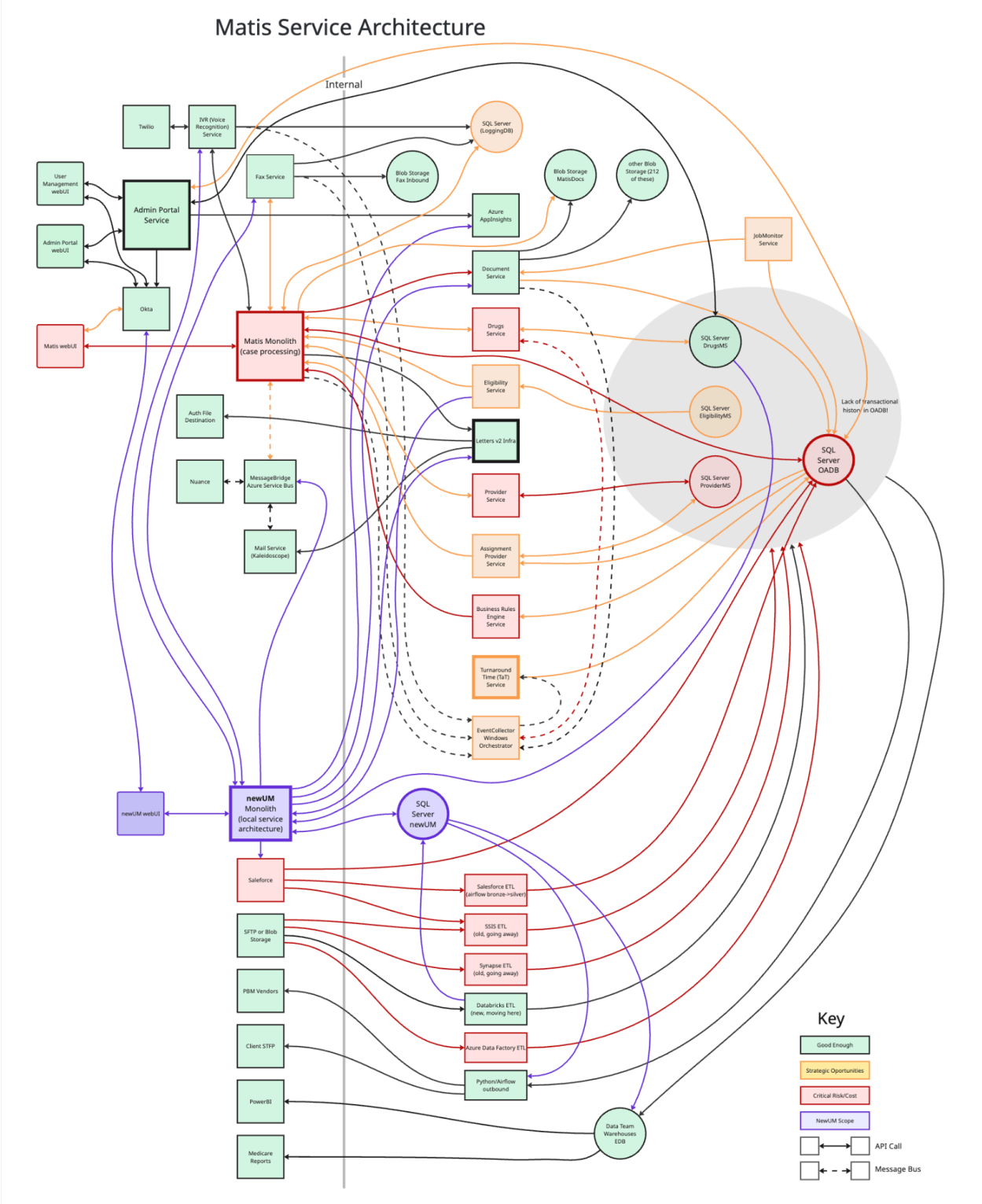


- **GOOD** - Services that can support the business without near-term attention.
    - **Admin Portal and DrugsMS** - Already integrated with SmartUM. The schema, functionality and technology are positioned to carry the business, if they can be integrated with Matis or its replacement.
    - **3rd Party integrations**, document stores, communication channels, authentication, and the like are serviceable.
    - **Letters V2**, is well considered, and has implemented service endpoints for each Matis custom case flow to *integrate with*. This has shielded the service from many of Matis' inefficiencies.
    - **D&A** - the Databricks and Airflow solution is good for reporting and good enough for ingest - until an Interop solution can be put into place. It is currently in operation for BCBSM, Geisinger, and Sentara, and there are plans to migrate three more by January, per provided documentation. D&A team has plans to migrate all clients to this stack, but are currently oversubscribed with operations load. There are a couple smaller projects that could help alleviate a portion of that load, addressed below.

- **BAD** - Services that currently are problematic, but not critical.
    - **Logging and Observability** services. Three systems are in place (Azure AppInsights, DataDog, and a SQL DB specifically for logs) but they need to be optimized and made available to all the development teams.
    - **Eligibility Service** is probably one of the most customized areas of the product. Does not have configuration capabilities. Custom intake pipes, etc.
    - **TaT** - The current project has built a solution with a logic core that is starting to catch up on compliance risks. Unlike Letters V2, it has deeply integrated with Matis and that portion of the code is problematic. There are redundant TaT calculations made on top of SmartUM and for reporting.
    - **Assignment Provider Service** - Logically stable, but has been having critical performance dropout issues, and should be instrumented and repaired.
    - **Drugs Service** - The Admin Portal implemented an adequate for drugs, but the database is accessed directly rather than using this service. This was updated significantly in support of the Monarch project but has not been fully integrated.

- **UGLY** - Services that are causing critical issues for OncoHealth.
    - **Matis Core** - as described above.
    - **Business Rules Engine -** Called out here because it is mostly bypassed with hard-coded case workflows, javascript code and UI behaviors. This would be replaced in a NewUM rewrite.

- **Provider Service** - manual Salesforce entry, multiple ETL scripts, insufficient data design. A new schema has been proposed but not implemented. Provider Mastering is not present.
- **Salesforce**, and the outdated D&A technologies (SSIS) cause redundant upkeep, operational load, and higher staffing levels.

2 - Paths Forward

Based on the observations made in the last report, we believe that the current teams are fully occupied with supporting the business, and have not had bandwidth to put significant efforts against improved product features or technical debt. The team has had significant attrition, and there are not viable succession plans for several staff functions.

Similarly, significant reduction in staffing would put the business at risk, either on compliance or meeting customer obligations. As new clients are added, the workload will scale linearly with QA, D&A, and Operational teams. There are some optimizations that are available, but these may only address critical gaps in team roles and skills.

Three scenarios for additional funding have been proposed with the intent of achieving some of the goals outlined. Below are the hypotheses that roughly fit the funding scenarios. The scope of work for this engagement has mapped the current state of OneUM, identified major risks and issues, and identified potential projects (some previously defined and some new). Additional research, planning and scoping will be needed to prove out and refine the hypotheses.

2.1 Scenario 1

Keep the lights on for 2-3 years.
12-Month Scope. $1M-$2M Incremental Spend

- Take on select higher-impact strategic projects, but leave Matis core and customization practices in place. Potential to address the following:
    - Matis UI - Time-boxed projects to repair *critical* gaps that drive operational cost and risk across the case lifecycle. Warning: many such gaps are not fixable due to Matis backend complexity. (see section 3.1 - Matis UI Critical Care - Medium)
    - Rudimentary Logging and Observability
    - Provider Schema and management (deprecate Salesforce), not including provider mastering. (See section 3.4 - Provider Management - Medium)
    - Deeper evaluation of Eligibility Service, Provider Mastering, and Interoperability projects. These are high-level pain points in the system that are out of scope for this scenario. Evaluations will help drive future product planning, and establish action plans if any of these areas become critical client issues.
    - Develop a business continuity plan, and off-ramp plan for SmartUM
    - Implement post-decisioning Turnaround Time functions (e.g., TaT requirements on appeals, and delegated appeals).

- Enable incremental D&A consolidation onto Databricks and Airflow, and retirement of Mirth
- Enable reorganization to support critically absent CS and Product disciplines.

This scenario will address a few of the critical scaling and compliance issues. It will address some of the team's grievances, and help address retention issues.

This *will not* address the root cause of OneUM's challenges. The baseline cost of R&D and Operations will continue to climb, and significant business and compliance risks will remain. This may not be enough to sustain the implementation demands and transaction volumes of a national payer.

2.2 Scenario 2

Address the root cause of OncoHealth's failure to scale. Addresses some but not all items in Scenario 1.

24-Month Scope. $4M-$8M estimated incremental spend

- Develop a NewUM monolith and database, featuring a case workflow engine, including a new UI that itself reacts to client-specific workflow configurations.
  - Develop this in-place within the OneUM service architecture.
  - Continuously migrate old data into the NewUM database as part of the SDLC.
  - To meet business goals this should be completed by mid-2026.
- Migrate Matis clients onto NewUM without "reimplementation"
  - Configurability and Migration will be included in the initial build.
  - Client-specific configuration documentation, configuration and extensive QA
  - Clients should be encouraged to receive materially-improved reporting. Extra effort for those that require legacy reports.
  - Implement all new clients on NewUM.
  - Goal to migrate all Matis clients by early-mid 2027, and decommission Matis.
- Develop a business continuity plan, and off-ramp plan for SmartUM.
  - Determine feasibility to migrate off of SmartUM and retire that as well.
- Above migration will significantly ease LOE for D&A to consolidate onto Databricks and Airflow.
  - Migrate certain clients to the new stack in the meantime.
- Enable reorganization to support critically absent CS and Product disciplines.

- Additionally, investigate and prototype AI-enabled optimization of PA, and real-time decisioning for a % of cases.

A proper scoping exercise is needed to validate and de-risk this scenario. Note that certain critical areas of development are not initially included in this scope. The goal is to enable the right-sizing of client implementations and D&A reporting, while both increasing efficiency and reducing the level of expertise (and cost) of case intake and decisioning. This should allow bandwidth for subsequent, proper strategic product development. That should include prompt attention to the remaining un-resolved strategic projects.

2.3 Scenario 3

Fully invest in OncoHealth's ability to scale and build value. Provides the best chance to expand the book, take on national payers, and expand into other areas of UM.
36-Month Scope. $5M-$10M incremental pend

- Fully address the identified major technical challenges and risks to growing OncoHealth's business. This includes all bullet points from Scenarios 1 & 2, above.
- More aggressive initial staffing to de-risk development, retiring legacy UM core platforms, and transitioning clients. Staff levels should be able to support parallel development of NewUM and select strategic projects.
- Addresses inefficient and more costly operations in a shorter time-frame. This improves margin on new business and reduces operational overhead.
- Allows for best efforts to introduce AI optimizations, that have promise to reduce costs of case intake (processing faxed cases), client data intake (e.g., leveraging AI capabilities in Databricks), and in increasing real-time PA decisioning. AI-enabled ventures have often reported multipliers on revenue and valuation.
- A few of the findings from this engagement have not been scoped in the prior scenarios. These can be prioritized here:
    - Provider Mastering (PDM). Research, scope and execution, on top of projects to adopt an improved provider schema and business logic.
    - Interoperability projects to create a more robust interface and integration service with payers, and to support CMS-0057 and CMS-9115 compliance.
    - Eligibility Service research, scope and development. Scalability, configurability, and potential AI optimization of client implementations.
    - Research on Member management or tracking "bellybuttons"
- Best chance to position the greater product development organizations to deliver strategic value, and expand opportunities for OncoHealth.

3 - Strategic Projects

The following sections detail *some of* the strategic projects that have been identified in our analysis and are mentioned in the above scenarios. There are other projects that are already well underway by OncoHealth including Letters V2, Turnaround Time improvements, etc.

None of these projects have yet been scoped by Newfire, and in most cases by OncoHealth. The following t-shirt sizing is provided for general reference against effort and risk.

Small < 1 team quarter
Medium ~ 2 team quarters
Large ~ 1 team year
XLarge > 1 team year

3.1 - Matis UI Critical Care - Medium

In lieu of a NewUM effort - we know there is a large backlog of improvements the case intake and clinical teams have identified in roughly three categories:

- The UI functionality is not correct and can be corrected.
- The UI does not have access to critical information in the greater OneUM systems, and forces Onco staff to log into multiple systems.
- The UI does not prevent inaccurate data or data gaps from being processed, creating downstream manual interventions by multiple teams.

A time-boxed project with the help from Newfire's UX team can address some of these areas and provide some relief and sense of directional improvement. Note that many of the real problems in the Matis UI stem from the underlying Drugs models and custom client workflows, and those would largely be out of scope for this initiative.

3.2 - Interop Infrastructure - Medium to Large

CMS Compliance, more efficient client data ingest and integrations

The key work items for OncoHealth to comply with CMS-0057-F are likely to include:

- Add the concept of "pending" interim statusing to OneUM PA workflow

- Build API to pass prior authorization status updates to payers immediately after each case status update and decision (not in end-of-day batches as is currently common), including comprehensive auth details such as units, codes, and requested duration
- Build configurability into case IDs used in outbound PA status updates to accommodate payers who cannot accept the same case ID twice (e.g. send “1234-pending,” then “1234”)
- Build mechanism to receive inbound auths from payers in standardized format (e.g. via X12 278 or other format, or potentially via manual intake initially), along with landing place in OneUM (e.g. inbound queue)
- Update contract/SOW language to define outputs/formats and responsibilities, and coordinate with each payer to confirm ingestion capabilities and any plan-specific constraints

The above work is expected to be a medium to large project. In order to produce a more accurate estimate and implementation plan, the following design and discovery must be done:

- Conduct inventory of current-state OncoHealth interfaces, file feeds, batch cadences, auth ID behaviors, exception cases (plans that can’t accept updates), and intake capabilities to determine what functionality gaps exist that would impact compliance (including gaps beyond those known today)
- Decide on API/data transmission specs (e.g. FHIR, X12 278, etc.), auth model (OAuth/SMART), and event timing expectations
- Determine the exact fields that must be part of case updates, including pending and final decisions, code systems involved, and any applicable reason codes and identifiers
- Evaluate project scope per payer customer, including which APIs/endpoints will be used, which lines of business are affected, expected volumes, and payer- or plan-specific requirements that deviate from the expected norm

3.3a - Data Back-End (Matis) - Small to Medium

The D&A team has settled on an effective data solution and strategy. Their challenge has been getting the bandwidth over the inflated workloads on compliance and reporting to address the cleanup of legacy solutions. Their client tracking chart, below, shows the level of redundancy, and that doesn't include some of the additional redundancies imposed by the two UM platforms (e.g., in calculating TaTs).

	UM Platform	Iris	Eligibility	Provider	Formulary	Claims	Med Auth	Med Resp	Rx Auth	Rx Resp	Notes
AHC	SmartUM		Synapse	Synapse	N/A	Prefect	AF / Interop	Airflow	N/A	N/A	
BCBSA	N/A	x	EDP	N/A	N/A	EDP	N/A	N/A	N/A	N/A	Jan 01 target
BCBSM	Matis	x	EDP	EDP	EDP	EDP	Airflow	Airflow	Airflow	Airflow	
BlueKC	Matis	x	EDP	EDP	EDP	EDP	Airflow	Airflow	Airflow	Airflow	Jan 01 target
BSWHP	Matis		Syn / ADF	Synapse	N/A	Prefect	Mirth	Airflow	N/A	N/A	
Cambia	Matis		EDP	EDP	EDP	EDP	Airflow	Airflow	Airflow	Airflow	Jan 01 target
Devoted	SmartUM		Synapse	Synapse	N/A	Prefect	AF / Interop	Airflow	N/A	N/A	
Evernorth	N/A	x	EDP	N/A	N/A	EDP	N/A	N/A	N/A	N/A	
Excellus	SmartUM		Synapse	Synapse	Synapse	Prefect	N/A	N/A	N/A	N/A	We currently process xls in smartum but are not delegated. Mar 01 target
FMHP	Matis		SSIS	Synapse	N/A	Prefect	N/A	N/A	N/A	N/A	
GEHA	SmartUM		Synapse	Synapse	N/A	Airflow	AF / Interop	Airflow	N/A	N/A	
Geisinger	Matis	x	EDP	EDP	EDP	EDP	Airflow	Airflow	Airflow	Airflow	
Highmark	Matis		ADF	SSIS	Synapse	Prefect	AF / Interop	Airflow	Mirth	Mirth	VT work on Elig. DE has begun provider
Humana	SmartUM		Synapse	N/A	N/A	N/A	AF / Interop	Airflow	N/A	N/A	
HP	Matis	x	Syn / ADF	Syn / ADF	N/A	Prefect	ADF	ADF	N/A	N/A	
MCS	Matis		SSIS	SSIS	N/A	Airflow	N/A	N/A	N/A	N/A	
MMM	Matis		SSIS	SSIS	N/A	Prefect	N/A	N/A	N/A	N/A	
P32	Matis	x	SSIS	SSIS	N/A	Old Dbx	Mirth	Airflow	N/A	N/A	Older jobs are commercial only. Medicare in progress. Move all to EDP 1/1
Sendero	SmartUM		Synapse	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Monthly elig job. Very low stakes
Sentara	Matis	x	EDP	EDP	EDP	EDP	Airflow	Airflow	Airflow	Airflow	
TripleS	Matis		SSIS	SSIS	N/A	Airflow	N/A	N/A	N/A	N/A	

Healthy	These processes are implemented into our new framework and running the way we would like
In Progress	We are in the process of migrating these to our new framework or doing the initial implementation
Refactor 1	Most critical refactor needs. Mirth is difficult to monitor, the code doesn't live in source control and is difficult to modify or debug
Refactor 2	Important refactor. Code is stable and we are able to monitor. However it is poorly written, inefficient and not easy to modify
Refactor 3	Backlog refactor. Some older airflow/dbx code we'd like to tweak and a dependency on an older SQL database we'd like to remove but we could live with this indefinitely

Adding resources to accelerate the porting from SSIS to Databricks (inside their Enterprise Data Platform - EDP) should accelerate the reduction of overhead, and risk of employee attrition and reporting compliance issues.

Note that under Scenario 2, and 3, that the migration onto EDP comes “for free”. See below.

3.3b - Data Back-End (NewUM) - N/A

Implementing NewUM (Scenarios 2,3) will move the data migration and normalization upstream into the NewUM OLTP. Since the D&A EDP (Databricks, etc) will be wired into the new systems the clients will inherit EDP by default. There will not need to be a separate migration effort.

The enhanced reporting capabilities under NewUM and EDP could be seen as an upgrade over current reports - but we know that can also be a point of friction with specific clients that demanded that reporting not change in their port from Matis to SmartUM. Some incremental effort would be needed to configure new reports back to their current formats.

3.4 - Provider Management - Medium

The proposed new Provider schema looks to be “good enough” either for Matis or SmartUM, and should be adopted after some additional review from a Newfire provider hierarchy expert. But the Provider workflow currently uses Salesforce CRM and requires a 3-step manual process. Building an Admin UI and new Provider Service could automate this, and remove large delays in cases with new providers.

Note that a related effort should be considered for the Assignment Provider Service - which has scaling and infrastructure challenges.

3.5 - Provider Mastering - Large

This is a placeholder for a project to provide some lightweight aspects of a Provider Data Management system. Currently OneUM treats each provider-payer relationship as unique. Further requirements analysis is needed here to better understand the tradeoffs.

Similar research is required to manage members and unique human beings (“bellybuttons”). Consideration should be given to commercial offerings for provider management, integration with payor provider directory APIs, and commercial platforms for member identity linking such as Verato.

3.6 - Eligibility Service Rework - Medium to Large

The Eligibility Service is relatively stand-alone, but not currently highly configurable. As this is an area that sees the highest level of client-specific customization, a focused refactoring effort here would accelerate implementations and reduce testing burden.

4 - NewUM Approach - XLarge

Our analysis work to date, including our prior readouts, indicates that the Matis monolith and the OADB are together the main block on OncoHealth's ability to achieve its goals. The complexity of the monolith continues to pile upon itself and has metastasized - the components around the monolith and its data store continually add complexity to deal with Matis.

"NewUM" would replace the monolith and several surrounding services with a simpler, configurable solution. We anticipate the following benefits from such an effort, as outlined in Scenario 2, above.

1. Reduces cost and complexity of implementations - lowers R&D load
2. Reduces liability and reputational risk from mishandled cases
3. Addresses case management inefficiencies, enables scaling
    a. Addresses missing data fields and key UI issues
    b. Case workflows can be configured by semi-technical personnel
    c. Opportunities to plug in AI rules
4. Reduces cost and complexity of unifying D&A tools
    a. Allows current D&A guardrails to shift left, preventing rather than catching errors
5. Enables retiring first Matis, and then SmartUM to simplify solution stack
6. Makes it possible to write code without making the system worse.

Overall, this project is required to unlock lower, more efficient operating costs and reduce risk to client relationships.

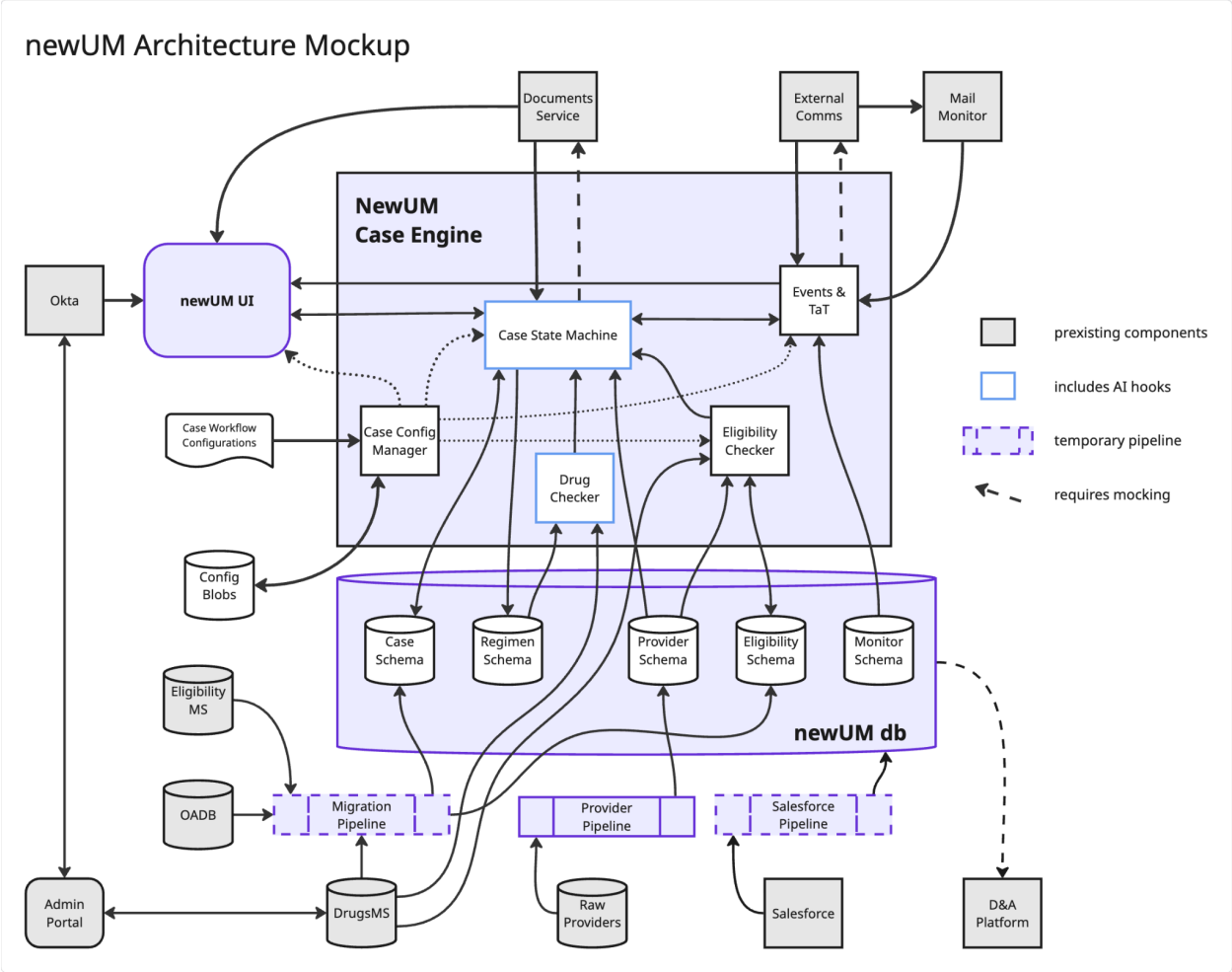
Architectural Approach

We have done an early, high-level investigation on what could be possible for a rewrite of Matis core, or NewUM. The following diagram illustrates what is possible. Please note the following:

- Much of the OneUM Service Architecture can support both Matis monolith and NewUM during and after development. These assets will not require rewrites solely in service of a new UM engine, although we would expect some upgrades may be required (including to align development processes with modern standards). They are represented in grey in the following diagram.
    - This includes Admin Portal and the Drugs DB (both recent efforts)

- In some cases mock interfaces would be used temporarily, to prevent write-back issues.
- The new UI would dynamically consume Case Engine configurations and react accordingly, without additional custom development.

- The new components would be built within the OneUM systems, in-place, with pipelines and an SDLC that supports Continuous Data Migration in addition to the usual Continuous Integration/Continuous Deployment (CI/CD). At completion, all member and client data will be live in the NewUM database.



Matis Client Migrations

To migrate clients from Matis to newUM involves a great deal of uncertainty. In order to prevent that uncertainty from becoming risk, we will do the following:

- Review prior customizations client-by-client
- Configure the new system to address these client-specific requirements

- Build temporary migration pipelines from the start of the project to force the new system to deal with the old data at every stage of development
- (in exceptional cases) Add additional, reusable configuration capabilities
- Extensively review test plans per client, comparing to initial implementation if possible
- Test continuously, supported by OncoHealth's QA team
- Review client reports - especially for clients now relying on the SSIS ingest pipelines
- Cut over each client when ready

When all clients are migrated off Matis, the total OneUM solution can be significantly simplified.

- Decommission Matis monolith, OADB, Matis UI, and several supporting services
- Clean up D&A scripts and tools
- Wind-down temporary migration pipelines

SmartUM Client Migrations

As mentioned in Section 1 - Migrating clients off SmartUM requires additional research, planning and collaboration with HealthFortis.

- It appears that migrations off of SmartUM (to either Matis or NewUM) may require some level of client participation. This needs further investigation, with the goal being to, at the most, require customers to send the same feed files to a different place.
- Data access agreements to support the migration strategy should be put in place.
- Business continuity provisions should be established, regardless of which scenario is pursued.

Once all remaining clients are migrated from SmartUM, the HeathFortis relationship could be terminated.

- Processes and tools established to manage SmartUM clients could then be decommissioned. This includes, but is not limited to TaT calculations, and D&A support (these would be covered under NewUM's solution).